Zhiheng (Kevin) Li

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EDUCATION

University of Minnesota, Twin Cities Minneapolis, MN

Bachelor of Science in Mathematics · Computer Science Minor *Expected May 2028*

- **GPA:** 4.0/4.0 | **Honors:** Dean's List, Fall 2025, Spring 2026 · Tau Sigma Honor Society
- **Relevant Coursework:** Calculus I, II & III, Linear Algebra & Differential Equations, Applied Linear Algebra, Proof-Based Sequences & Series, Physics for Scientists and Engineers I & II, Programming in Python and Java, Introduction to Algorithms & Data Structures, Introduction to Statistical Analysis, Cryptology & Number Theory
- **Directed Reading Program (Fall 2026):** *Measure and Integral: An Introduction to Real Analysis* (Wheeden & Zygmund, 2nd ed.), as foundations for Optimal Transport, with mentor Anisha Jain, Ph.D. student in Mathematics, University of Minnesota

RESEARCH EXPERIENCE

SuperCDMS Dark Matter Experiment - School of Physics & Astronomy, University of Minnesota Minneapolis, MN

SuperCDMS Researcher & Research Assistant to Prof. Yan Liu, UROP Fellow *Oct 2025 - Present*

- Built **cdms_spotlight**, a multilingual natural-language query system for the SuperCDMS DQM database hosted at **SLAC**: server-side filtering scaling from ~1,300 to ~100,000 series, cuts on any of 10,929 ODB parameters per series, self-learning local NLP (TF-IDF + logistic regression); runs continuously as a Slack bot in the official collaboration workspace, pip-installable CLI; **presented at the SuperCDMS Software Meeting on July 14, 2026**; contributed to the follow-up DQM/ODB upgrade, prompted by cdms_spotlight, that moves non-metadata parameter cuts server-side so queries no longer pull data to the client. Jul 2026

SNOLAB Phonon Pulse Analysis (Run 4, Run 7) May 2026 - Present

- Built the **phonon pulse template pipeline** for 13 SNOLAB R4 detectors (12 phonon channels each): event selection around the 10.37 keV activation line (~120 GB raw traces), free-pretrigger 2-exponential fits with quality cuts; the 1×1 and PCA N×M ROOT templates serve as the R4 pulse templates for optimal-filter reprocessing; **presented at the SuperCDMS Analysis Working Group meeting on July 29, 2026**. May–Jul 2026
- Measured the **phonon collection efficiency** of detector Z7: converted every K-line pulse from current to power with the TES electrothermal equations and integrated the fitted pulse in closed form, validated against the official absorbed-energy RQ; **presented at the SuperCDMS Analysis Working Group meeting on September 23, 2026**; the current-to-power step uses the instant-sensor, infinite-feedback limit, refined below. Jul–Sep 2026
- **Full-model collection efficiency** of detector Z7: the efficiency above assumes an instant sensor with infinite feedback; derived the two coupled electro-thermal equations, linearised them to the complex impedance and fitted its four unknowns (current sensitivity, loop gain, thermal time constant, inductance) on Run 7 data with code written from the equations, reproducing the Irwin and Hilton (2005) two-time-constant response; the data show a **third pole, still an open question in the collaboration**; tested a second thermal body against the readout electronics, on the data and with the readout board measured alone on the bench, and placed it in the sensor; next, data from a second injection point, then a simulation to estimate the efficiency of low-energy events. Oct 2026 - Present
- Compared every **energy estimator** the Run 7 RQs allow on the same K-line events of Z7; a linear regression on the per-channel optimal-filter amplitudes gives several times the resolution of the plain OF sum. Oct 2026

Lab Instrumentation Oct 2025 - Jul 2026

- Built **BlueFors-Alert**: minute-level PostgreSQL sync of the 64 GB CS2 database to a Raspberry Pi; Slack monitor with dual IDLE/COLD thresholds and time-range plots of any sensor; extended to a liquid-nitrogen scale. Jul 2026
- Extended RuOx (R31279 & R31839) **calibration curves** to 5 mK–300 K (PCHIP + Mott VRH); the generated .340 files are now in use on the lab's BlueFors dilution refrigerator. May 2026
- Built a **Python pipeline** for LED calibration data from 11 cryogenic Ge channels (5,200+ events, 10+ GB raw): baseline subtraction, SNR-based glitch detection, peak alignment, exponential fitting. Oct 2025

Mouse Eye Tracking for Neural Alignment - Deisseroth & Mitra Labs, Stanford University Remote

Research Collaborator (remote), with Kaiwen Sheng, Ph.D. student in Bioengineering *Aug 2026 - Present*

To align eye behaviour with brain recordings, developed a pipeline that, for most videos, needs only 20–40 human-labeled frames to predict every one of a video's ~70,000 frames accurately.

- Set up **DeepLabCut** tracking of eight eye keypoints (pupil ellipse, eyelids, eye corners) in infrared videos recorded over multiple days; fixed the training recipe by controlled experiments on held-out test frames (batch size, ResNet-50 among five backbones, the "jump" frame-picking rule) and defined every keypoint precisely. Aug–Sep 2026
- Ran **active learning on 19 videos from 3 mice** to find how many labels each video needs (first 20 frames by k-means, then 20 per step where tracking is unstable, until two steps improve the error by under 3 %): most videos saturate at

20–40 labels; a new mouse needs its own labels (208 px to 10.4 px with 20) but starts better with every mouse seen; adding videos never degraded earlier ones; 5 or 10 labels cannot replace 20. Sep–Oct 2026

- Compared nine algorithms for the **pupil area** (four-point ellipse closest to the human labels) and found a simple, accurate **per-frame eye-opening criterion**: the eyelid distance, with each mouse's threshold the best of all tried on 327 frames judged by eye (94 % and 89 % accurate); both unsmoothed, for alignment with brain recordings. Oct 2026
- Designed a **self-correcting training loop** for the ~2 % of 1.3 M frames still wrong after labeling saturates, too many to check by hand and not to be smoothed over time: it **finds wrong frames without labels**, **repairs them from the network's own heatmaps** under a geometric check from the human labels, and **retrains on them**. **Five controlled experiments** showed that unflagged and repaired frames both improve the model and chose the repair and closed-eye rules; every threshold is calibrated on labels or frames judged by eye. Loop now running; once it saturates, will analyse whether the error can be reduced further. Oct 2026 - Present

RESEARCH & INDUSTRY EXPERIENCE

Low-Carbon Transition & Industrial Monitoring Platform Beijing / Shanghai, China

Faculty-led Researcher (research phase) → Algorithm Engineer (industry phase), Shanghai Huaruizhongxin Technology Co., Ltd.

Oct 2024 - May 2025; Aug 2026 - Sep 2026

- Contributed to a faculty-led low-carbon transition project for China's steel industry, supporting a proposal and implementation plan for national ministry consideration through data collection, database construction, and quantitative analysis.
- During the industry implementation phase, contributed to an energy and carbon-emissions monitoring platform by designing an algorithm prototype for industrial time-series data quality governance, translating meter/point metadata and quality flags into rules for cumulative, instantaneous, consumption, enumeration, and statistical measurements.
- Implemented historical and real-time cleaning for range, constant, trend, missing-data, and communication anomalies; used same-time historical profiles for outage imputation and cumulative-to-consumption attribution; validated multi-day data with plots and regression tests, then packaged and delivered the work.

SELECTED PROJECTS

College Students Entrepreneurial Project & Entrepreneurship Roadshow Beijing, China

Team Leader *Oct 2024 - Jun 2025*

- Led a student team to design a **billiards rest** that combines the high and low bridge heads into one;
- Took it from CAD model to 3D-printed prototypes, iterating through player testing and market research; received **national project approval with seed funding**; **Chinese utility patent** under CNIPA review.
- Invited to present at the **Tsinghua X-Lab & Kering Entrepreneurship Roadshow** to industry leaders.

AWARDS & HONORS

- **UROP** (Undergraduate Research Opportunities Program) Fellowship, University of Minnesota, Summer 2026
- **1st Prize**, Beijing University Student Innovation & Entrepreneurship Competition (Science & Technology Track)
- **1st Prize**, National College E-Commerce “Innovation, Creativity & Entrepreneurship” Challenge, Campus Level | Team Leader
- **3rd Prize**, China International College Students’ Innovation Competition, Campus Level (2025) | Team Leader
- **2nd Prize**, NECCS National English Competition for College Students

ACTIVITIES & SERVICE

- **Special Olympics USA Games Minnesota 2026**, Tennis Division - Volunteer Ball Retriever | June 25, 2026
- **Chinese Student Union**, UMN - Active Member | Oct 2025 - Present
- **Chinese Students & Scholars Association**, UMN - Volunteer Calculus I Tutor | 2025 - Present
- **College Table Tennis Club / Football Club / Guitar Club** - Team Member | 2023 - 2024

PROFESSIONAL SKILLS

- **Techniques:** Cryogenic instrumentation, signal processing, data pipeline development, thermometer calibration curve fitting, real-time PostgreSQL sync, local NLP, Slack bot development, waveform template construction, TES / SQUID readout modelling, electronics transfer-function measurement, MIDAS DAQ control, keypoint tracking, active learning
- **Tools:** Python (NumPy, Pandas, SciPy, Matplotlib), C++, scikit-learn, DeepLabCut, PyTorch, PostgreSQL, Java, ROOT, MIDAS / ODB, Git, Linux, Slurm, MSI HPC (Agate)
- **Languages:** Mandarin Chinese (Native), English (Fluent; IELTS 7.0)